

**HSLU**

**AI/ML**

# **NeurIPS - Ariel Data Challenge 2025**

Kaggle Challenge

AI Competition Report

## **Team**

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# Declaration of Originality

We hereby confirm that we are the sole authors of the written work here enclosed and that we have compiled it in our own words. Parts excepted are corrections of form and content by the supervisor. We have stated all sources, literature and other aids used, have specified passages taken literally or in terms of content as such, will protect the client's interest in confidentiality and will respect the copyright regulations of the Lucerne University of Applied Sciences.



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## **Abstract**

The ARIEL mission, scheduled to launch in 2029, will characterize the atmospheres of 1 000 exoplanets through transit spectroscopy. Extracting meaningful atmospheric signatures from observations requires robust methods to handle systematic noise, detector artefacts, and the inherent challenge on distinguishing faint planetary signals from stellar variability. This project addresses the 2025 ARIEL Data Challenge, which tasked participants for the 6th year in a row, with developing algorithms capable of predicting the transit depth spectra across 283 wavelengths from simulated observations, including both point estimates and associated uncertainties.

Approaches of other participants were explored, as well as a de-jittering algorithm. Following an existing preprocessing pipeline, a multi-headed Convolutional Neural Network (CNN) and three variations of it were developed to process parallel data streams from two sensors and stellar parameter inputs. The best model achieved a competition score of 0.283 on Kaggle.

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# 1. Introduction

The Atmospheric Remote-sensing Infrared Exoplanet Large-survey (ARIEL) mission is part of the Cosmic Vision program, asking "*how do planets form and what are the conditions that might make them (or their moons) habitable*" (Tinetti et al., 2021, p. 13). Realistic progress can now be made to answer the origin of life in the universe, by understanding exoplanets, their formation, evolution and condition for habitability (Yip et al., 2024, p. 1). Exoplanets are planets outside of our Solar System (see Figure 1.1) (Yip et al., 2024, p. 1). Their study contributes to form a better understanding of habitability, by helping to clarify the place of our Solar System and the Earth in the galactic context (Tinetti et al., 2021, p. 13). Since their discovery in the 1990s, (Tinetti et al., 2021, p. 12), over 5 600 exoplanets were discovered (Yip et al., 2024, p. 1). Different methods exist to identify them, such as transit, direct imaging (see Figure 1.1) and radial velocity (Agency, n.d.). Their mass, sizes and orbits differ broadly; they can be rocky and similar to the Earth or large gas giants (Tinetti et al., 2021, p. 10). No pattern is known between the planet's characteristics, such as its size, orbital parameters, and its host star, and there is much to discover about their surface and atmosphere, which role they take, and the influence of their host star remain to be explained (Tinetti et al., 2021, p. 10).

In 2029, ARIEL (see Figure A.1 in appendix) will be launched to study 1 000 exoplanets or *extrasolar planet* in our galactic neighborhood (Yip et al., 2024, p. 1). While it will collect high-quality, high-precision data on a large number of exoplanets, novel approaches are required to face unsolved machine learning problems (Yip et al., 2024, p. 3). Therefore, the largest planetary science conference of the world, the European Planetary Science Congress launched the Ariel Data Challenge Award to encourage interdisciplinary collaborations between the fields of astrophysics and machine learning (Yip et al., 2024, p. 3).

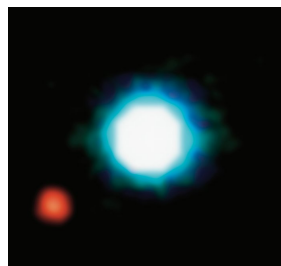


Figure 1.1.: In 2004, *2M1207b* (the red point on the left), became the first exoplanet to be directly imaged. From ESO. (2010). *2M1207b* - first image of an exoplanet [[https://www.eso.org/public/images/26a\\_big-vlt/](https://www.eso.org/public/images/26a_big-vlt/)].

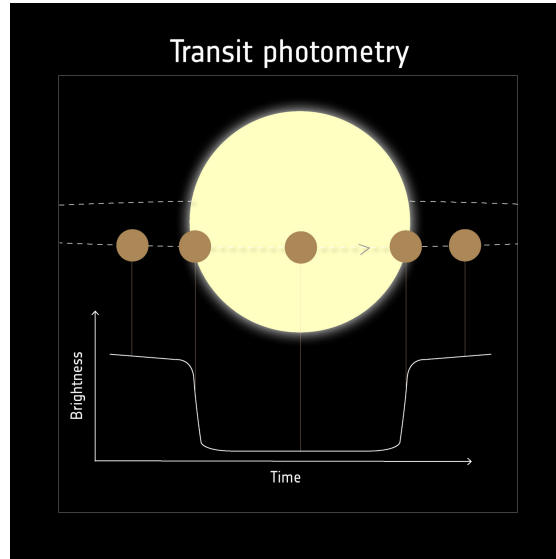


Figure 1.2.: Illustration of the transit method used to discover exoplanets. From ESO. (2010). 2M1207b - first image of an exoplanet [[https://www.eso.org/public/images/26a\\_big-vlt/](https://www.eso.org/public/images/26a_big-vlt/)].

## 1.1. Problem statement

To advance exoplanet areas regarding atmospheres, composition, formation, and evolution, the spectroscopic survey made by ARIEL will be key, as it will bring information about exoplanets of multiple sizes, temperatures and stellar types (Tinetti et al., 2021, p. 19). One of the hardest data analysis problems of astronomy is to extract faint signals about exoplanets from noisy observations (Yip et al., 2024, p. 1). When an extrasolar planet transits in front of its star, a fraction of the starlight (20-100 photons per million) crosses the exoplanet's atmosphere (see Figure 1.2) (Yip et al., 2024, pp. 1-2). It interacts with its chemistry, clouds, and winds and is between 50 and 200ppms (Yip et al., 2024, pp. 1-2). Knowing the atmospheric composition of these exoplanets would help discriminate their nature (Tinetti et al., 2021, p. 13). These signals are captured by telescopes like ARIEL, which provide the spectroscopic data. However, they also corrupt it with *jitter noise*, due to the spacecraft vibrations and drift (Yip et al., 2024, pp. 1-2) (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 2). The jitter noise is often compared with taking a long-exposure image with a shaky hand (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025; Yip et al., 2024, p. 2). It adds excess noise above photon level in the short wavelength channels and must be corrected (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025; Yip et al., 2024, p. 2). Other types of noises are present in the signal, such as detector noise, thermal noise and astrophysical noise, as mentioned by (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025).

### 1.1.1. Background and Motivation of the Problem

Ariel will be able to distinguish the nature of tens of transitional planets by estimating the mean atmospheric molecular weight and, for a representative sample of them, by performing an in-depth atmospheric characterization, also aimed at directly detecting the primordial hydrogen through observing features of H-rich ionic species (Tinetti et al., 2021, p. 34).

The Ariel Data Challenge aims to create synergies between the machine learning and the astrophysics community (Yip et al., 2024, p. 3). With the help of simulated data, these challenges support the development of advanced algorithms and introduce new techniques to prepare for the coming mission. Its particularity lies in the fact that no ground truth exists; the mission is exploratory and will gather unseen data. Thus, the algorithms must be robust to outliers. This challenge revolves around the chemical spectrum of the exoplanets' atmospheres (University College London, 2025a). Planets, unlike rocks, have an atmosphere. This is visible when observing the planet transiting: some light of the star shines through the planet's atmosphere or *gaseous envelope*. At that moment, a dimming of the stellar flux passes between the observer and the star (Nikolaou et al., 2023, p. 700). The transit is visible in the stellar flux time series as a dip, corresponding to the ratio of the planet to stellar radius (Nikolaou et al., 2023, p. 700). The absorption of the specific wavelengths is observed in the stellar flux loss and signatures, such as  $H_2O$ ,  $CO$  and  $CH_4$  - critical to foresee life - are identifiable (see Figure A.3 in appendix) (Nikolaou et al., 2023, p. 700). The goal of the challenge is to predict the mean and variance of all instances (University College London, 2025a). The data can be manipulated in different domains: temporal, spatial and spectral (Yip et al., 2024, p. 4). To better understand the use of the challenge's output, the paper of Forestano et al. (2025) can be consulted. In it, they present a range of supervised machine learning methods that could exploit the mean and uncertainty for atmospheric retrievals.

## 1.2. Goal of this Project

In the context of the module *AI Competition HS 25*, we participated in the Kaggle Competition NeurIPS Ariel Data Challenge 2025<sup>1</sup>. While the official competition was closed on the 18<sup>th</sup> of September, corresponding to the beginning of our semester, it remained possible to submit notebooks and receive their corresponding scores (Yip et al., 2024, p. 1). Since participants' notebooks and research papers were available, they were reviewed (see Chapter 2) and the data was analyzed (see Chapter 3). A CNN model was developed as baseline, improved and complemented with Recurrent Neural Network (RNN) or Temporal Convolutional Network (TCN) (see Chapter 4). Training optimization was done with WeightsAndBiases<sup>2</sup> and best models were submitted (see Chapter 4). A final score of 0.283 was achieved, which is analyzed in Chapter 4.

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<sup>1</sup><https://www.kaggle.com/competitions/ariel-data-challenge-2025>

<sup>2</sup><https://wandb.ai/>



## 2. State of the Art and Machine Learning Theory

Searching for suitable approaches and models, two axes were investigated: Kaggle notebooks from the competition, and scientific papers. The main findings will be presented in this chapter. Research papers discussing astronomy and unrelated to machine learning were left out.

### 2.1. Lessons learned from the 1st Ariel Machine Learning Challenge: Correcting transiting exoplanet light curves for stellar spots

Nikolaou et al. (2023) focus on automating the derivation of transit depths with efficiency and precision, despite the presence of stellar spot. In the paper, they present the work of the first five winners of the first ARIEL challenge in 2019.

The star-spot crossing problem comes from a combination of challenges that can be handled by machine learning. Nikolaou et al. (2023) mention the large amount of data to process, the low Signal-to-Noise Ratio (SNR), non-linear patterns with unknown parametric form, the different forms of available information, such as time dependent and independent, and a high degree of degeneracy (Nikolaou et al., 2023, p. 696). The data set used was simulated and contained 55 noisy light curves, represented by 300 time-steps corresponding to a 5 hour observation window by ARIEL. In addition, the following parameters were at disposal: orbital period, stellar temperature, stellar surface gravity, stellar radius, stellar mass, and stellar K magnitude (Nikolaou et al., 2023, p. 700). Both training and test sets had no common data point nor common points from the same planet (Nikolaou et al., 2023, p. 700). They defined this problem as a multi-target regression task (Nikolaou et al., 2023, p. 700). Nikolaou et al. (2023) proceeded to present the solution of the first five winners.

It must be noted that the first challenge, while on the same topic as the 2025 challenge, differed in terms of data and prediction. The goal of the challenge was to recover the radius from light curves altered by stellar spots (Nikolaou et al., 2023, p. 702).

Their baseline model was a fully connected deep neural network run over 5 epochs with no early stopping and achieved a score of 8 726 out of a possible 10 000.

James M. Dawson, an Astrophysics PhD student at Cardiff University, won the challenge with a score of 9 813 (Nikolaou et al., 2023, p. 702, 705). He presented a 1D CNN, as the 2D gave worse results (Nikolaou et al., 2023, p. 703). Flux were normalized and clipped when necessary. The architecture contained 3 1D CNN and 6 linear layers. Average pooling and ReLU were used in between. The planetary parameters were only introduced in the last

linear layer (Nikolaou et al., 2023, p. 703). The training took place over 75 epochs with a batch of 256 and Mean Squared Error (MSE) for the loss function. No overfitting was found and no additional normalization technique, such as batch normalization, was used (Nikolaou et al., 2023, p. 703).

The second winner team came from the field of machine learning and data science, at the University of Tübingen in Germany (Nikolaou et al., 2023, p. 703). Their aim was to ensure robustness to outliers. Therefore they applied no preprocessing, except for rescaling the features and targets. Their proposed architecture consisted of two branches; one for the light curves, the other one for the stellar and planetary parameters. The light curves passed through two fully connected layers, a batch normalization and dropout before being concatenated with the output of the fully connected layer applied on the stellar and planetary parameters. After the concatenation, the data passed through a sequence of fully connected layer, batch normalization, dropout and fully connected layer outputting 55 neurons mapping the predicted relative radius (rescaled) (Nikolaou et al., 2023, p. 704). They ran 1 000 epochs and a patience of 20 over a batch of 3 048, using MSE for loss function. A 10-fold cross-validation was used and the average of the outputs of all 10 models were calculated for the final prediction. They reached a score of 9 812 (Nikolaou et al., 2023, p. 702).

The third place was also held by data scientists from the Jožef Stefan Institute and Bias Variance Labs. Their preprocessing was heavier than with the other teams and they represented the stellar spots of the star-planet pair as Gaussian noise instances to decrease the noise of the data (Nikolaou et al., 2023, p. 704). In addition, they aggregated several measurements to reduce the variance (Nikolaou et al., 2023, p. 704). Using Planck's law, they estimated the energy emitted by the stars (Nikolaou et al., 2023, p. 704). Their model consists of three separate models: a random forest tree (500 trees), an extreme gradient boosting (150 trees), and a multi-target fully connected neural network with a hidden layer of 100 neurons (Nikolaou et al., 2023, p. 705). Their model was trained with batch normalization, dropout and ReLU for the activation function. It trained over 1 000 epochs and used MSE for loss function (Nikolaou et al., 2023, p. 705). They optimized their model's weights manually and finally achieved a score of 9 808 (Nikolaou et al., 2023, pp. 702, 705).

The fourth place was held by a group of researchers and data scientists from the Space Research Institute (Austria), Know-Center (Austria), and the University of Passau (Germany) (Nikolaou et al., 2023, p. 705). Using a linear regression, a light and interpretable model, they reached the fourth place with a score of 9 805 (Nikolaou et al., 2023, p. 702). To do so, they reduced the photon noise and normalized the light curves using percentiles, resulting in estimates of the differences of the dip. These were used in multiple linear regressions. They had one model per wavelength. Their only hyperparameters was thus these percentiles, which they found manually and iteratively (Nikolaou et al., 2023, p. 706).

Finally, the last presented team came from the TU Dortmund and was made up of machine learning researchers working in astro-particle physics (Nikolaou et al., 2023, p. 706). During the preprocessing, they used z-scale piecewise aggregate approximations to simplify the input. They trained a fully connected Deep Neural Network (DNN) on these extracted features as well as the stellar and planetary parameters (concatenated) over three layers of dense layers.

A short analysis is provided by the authors at the end of the paper, and pointed out the

flexibility of machine learning techniques. All five scores were very similar while the used models and their preprocessing differed broadly. The critical point was to address outliers (Nikolaou et al., 2023, p. 707). Indeed, all solutions *“either construct highly non-linear (w.r.t. the raw data) models with minimal preprocessing – deep neural networks and ensemble methods – or amount to obtaining meaningful statistics from the light curves, constructing linear models which yield comparably good predictive performance”* (Nikolaou et al., 2023, p. 695).

This paper provided a broad overview of machine learning techniques used in the past challenges. While the predictions differed, the reasoning of the participants remained of high interest. Surprisingly, the models did not rely on big modules like ResNet but rather on simple and customized models.

## 2.2. De-jittering Ariel: An optimized algorithm

Jitter noise is a source of disturbance of the signal. Anticipating ARIEL’s mission, Bocchieri, A., Mugnai, L.V., Pascale, E. et al. (2025) proposed a detrending algorithm, based on spatial information. In their paper, they detailed how they generated their own dataset with Exoplanet Observation Simulator (exosim), extracted the signal and detrended it (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025). In addition, they carefully analyzed their results. An attempt to replicate their algorithm is presented in Section 4.1.1. The authors exploited the characteristics of ARIEL design and focus on minimizing jitter disturbances relative to the photon noise at all timescales. Their approach aims to be robust to non-stationary jitter noise and avoid flat-field calibration (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 6). Their algorithm and analysis is processed on only one of ARIEL’s sensor (FGS-1) for simplicity (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 6). Inspired by detrending methods for post-cryogenic, their approach exploits spatial information on Point Spread Function (PSF) distortion to detrend the optical signals (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, pp. 5-6). After the data generation and the signal extraction, they described the PSF and its effect, leading to the explanation of their algorithm. Considering the jitter adds photometric noise by shifting and changing the PSF. Thus, the optical signal is sampled with different focal planes (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 10). For that reason, Bocchieri, A., Mugnai, L.V., Pascale, E. et al. (2025) tried to correlate the shifts and distortions in the PSF (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 10). They describe the sampled signal versus the incident signal as follows:  $S_\lambda(t) = f_\lambda(\vec{X}) \times \phi_\lambda + n_\lambda(t)$  where,  $S_\lambda(t)$ ,  $\phi_\lambda$  and  $n_\lambda(t)$  correspond to the sampled signal, incoming optical signal and noise at wavelength  $\lambda$  and time  $t$  (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 10). The unknown wavelength-dependent function is  $f_\lambda$ . The low-order moments of light distribution on the focal plane in photometric and spectral images determined the distortion parameters of the parameter  $\vec{X}$  (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 10). The position of  $\vec{X}$  ( $x$  and  $y$ ) was its shift, and its width was the PSF spread ( $w_x$  and  $w_y$ ). The skewness ( $s_x$  and  $s_y$ ) and kurtosis ( $k_x$  and  $k_y$ ) were the asymmetry and tails of the distribution, indicating the PSF distortion (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 10).

Their algorithm consisted of four steps:

1. Estimation of position, width, skewness and kurtosis.
2. Registration of spectral images on the same grid as the estimated moments ( $x$  and  $y$ ).
3. Correlation of estimated parameters with measured signal
4. Suppression of identified correlations from the measured signal (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 11).

During the transit, the formula includes the planet's parameters: radius, semi-major axis, orbital inclination, eccentricity inclination, time at mid transit and period (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 11).

$$S_\lambda(t) = f_\lambda(\vec{X}) \times \phi_\lambda \times \Lambda_\lambda(\vec{P}, t) + n_\lambda(t)$$

They estimated both the parameters of the planet and the nuisance parameters  $c_k$  by fitting the model to each photometric and spectroscopic timeline (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 11). The estimation of low-order moments of the focal plane were estimated along pixel rows and columns of the focal plane image, centered around an estimate of the centroid obtained with the barycenter method and improved later with Iterative Weighted Center of Gravity (IWCoG) (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 11). The centroid estimation is repeated until convergence. A Gaussian was used in the weighting function its widening factor was found empirically. It was centered on the centroid estimation (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 11). The moments were estimated from the five brightest spectral bins, as the jitter disturbance is correlated across all spectral bins (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 12). Finally, the model is fit on photometric and spectroscopic light curves with a non-linear least square method from the *SciPy* library (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 12). In more simpler terms, the planet parameters defined the expected signal, and the dejitter algorithm removed noise around that signal so the parameters can be estimated more accurately.

Later, the authors proceeded to do a noise analysis on raw, detrended, and reference signals (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 12). The Allan deviation was used to evaluate the amplitude of the noise at different time scales (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 12). A significant improvement was demonstrated (Bocchieri, A., Mugnai, L.V., Pascale, E. et al., 2025, p. 19).

## 2.3. Kaggle notebooks 2025

The first-place solution ((Jeroen Cottaar, 2025a, 2025b, 2025c)) processed raw sensor data counts from Ariel Infra-Red Spectrometer (AIRS) (binning 5) and Fine Guidance System (FGS) (binning 50), applying cosmic ray removal to compute amplitude signals  $S(\lambda, t)$ . A Bayesian prior decomposed the signal into multiple components: uncorrelated Gaussian noise, and unregularized stellar spectrum (283 degrees of freedom), polynomial drift (third

order), transit window parameters modeled with BATMAN (11 parameters, including limb-darkening and orbital elements), and transit depth variations. The transit depth variation itself was decomposed into three components: a single Gaussian values (FGS), a Gaussian process with two squared-exponential kernels with 282 DOF (AIRS), and five Principal Component Analysis (PCA)-derived basis functions. The non-linear model was solved through iterative linearization, initialized via grid search over transit depth and mid-time, followed by FGS optimization of drift parameters, and finally eight iterations of Bayesian Inference. From the winner's detailed write up it is clear, that the solution depended critically on "fudging" - empirical post-hoc corrections to both transit depth means and uncertainties per sensor, optimized on training data. This contributed 0.152 to the final score of 0.648. The author explicitly stated this to indicate a fundamental gap in the physical model, likely the transit modeling itself. This, combined with the domain-knowledge heavy preprocessing pipeline made this solution unsuitable for our learning objectives.

The second-place solution took a fundamentally different approach, relying on classical feature engineering rather than physical modeling. They extracted transit characteristics by analyzing signal derivatives to find boundaries, then computed hand-crafted features describing transit shape and depth. Two Ridge Regression models handles normal versus outlier cases, with polynomial detrending to remove baseline drift. Uncertainty estimates came from bootstrap resampling across 100 model iterations combined with cross-validation residuals. This approach proved competitive, but contained similar to the first place a key limitation: The features and thresholds required carefully manual tuning. The separation of outliers at frames 20 - 354, the choice of specific percentiles and slope metrics, and the weighting scheme for uncertainty (75% bootstrap, 25% RMSE), were design decisions optimized for the training data and therefore failed to generalize on the hidden test set and resulted in a score of 0.612. (BEEMAX, 2025b).

The third-place solution combined CNN-based feature extraction with a rational quadratic neural network for regression. Transit phases were identified by computing gradients of moving-averaged signal to locate ingress/egress boundaries, similar to the second-place approach. Unlike other solutions, no binning was applied, instead AIRS spectra were sliced into 32 wavelength slices, and features extracted through sliding windows. These spatial features were fed into a multi-layer CNN architecture. The regression head employed a rational quadratic basis function network, computing the distances between input features and learned anchor points. The architecture directly predicted both transit depth means and log-variances for uncertainty estimation, resulting in a score of 0.603 (BEEMAX, 2025a; MoonPole, 2025). (University College London, 2025b).

### **2.3.1. General Learning**

The analysis of participant notebooks beyond the top three placements reveals heavy reliance on the preprocessing pipeline provided by the competition organizers. Consistent with the findings from the 2019 challenge Nikolaou et al. (2023), the highest performing solutions employed custom architectures and relatively simple models, rather than complex end-to-end deep learning approaches. Notable, despite Yip et al. (2024) anticipating that winning solutions would integrate information across temporal, spatial, and spectro-spatial

domains, the submitted notebooks showed little evidence of such comprehensive multi-domain model fusion. This pattern underscores that the challenge fundamentally required deep understanding of the underlying physics: the complex interactions between planetary atmospheres, stellar radiation, and instrumental response. Success depended less on sophisticated neural architectures than on correctly modeling the physical phenomena governing exoplanet transit spectroscopy and the systematic effects inherent to space-based observations.

## 3. Data Quality Assessment

The data provided by the challenge-issuer European Space Agency (ESA) comprises 1 100 synthetic planetary observations. These signals were generated using the ExoSim2-library (Mugnai, Lorenzo V. and Bocchieri, Andrea and Pascale, Enzo and Lorenzani, Andrea and Papageorgiou, Andreas, 2025) developed by the ARIEL Consortium and S2MD group specifically for the upcoming ARIEL space mission<sup>1</sup>. ExoSim2 is a comprehensive observation simulator that models the entire detection pipeline from focal plane creation through radiometric modeling to final Non-Destructive Readout (NDR) generation and was designed to imitate the instrumental systematics, noise characteristics, and observational artifacts expected from the physical ARIEL instrumentation.

It should be noted that this challenge has been issued yearly since 2019 with increasing complexity concerning the data structures. The 2025 challenge introduces three key modifications from prior iterations. First, each planet has a unique host star with distinct mass, radius, and temperature, eliminating parameter degeneracies present when all targets share identical stellar properties. Second, repeated observations per target enable variance quantification across independent measurements and introduce observation-to-observation systematics. Third, the added limb darkening - the wavelength dependent intensity variation across the stellar disk - adds 1-5% photometric structure to transit profiles, complicating flat-fielding and baseline subtraction during calibration (see Figure A.4 in appendix).

### 3.1. Data Source

The ARIEL probe will have two sensors that capture planetary observations during the observation window of 3.75 hours. The FGS - channel 1 which was designed for high precision photometry. It captures 135 000 images per observation on the wavelength spectrum between 0.6 and 0.8  $\mu m$  (see Figure A.2) which is on the spectrum visible to the human eye. As the name suggests this sensor is primarily used for guidance and orientation of the probe itself. The AIRS - channel 0 captures the mid range of the infrared spectrum between 1.95 and 3.9  $\mu m$  at a rate of 11 250 images per observation. This sensor has a much broader capture range and will contribute 282 out of 283 wavelengths to the submission.

#### 3.1.1. Signal Data

Each planetary observation identifiable by a unique planet ID is provided in the following format:

---

<sup>1</sup><https://exosim2-public.readthedocs.io/en/latest/>

FGS1\_signal\_[observation\_number].parquet:  
 Shape: (135000, 1024) (135000 frames × 1024 pixels)  
 32×32 detector flattened to 1024 pixels  
 Data: uint16 ADC counts, range 330-42172

AIRS-CH0\_signal\_[observation\_number].parquet:  
 Shape: (11250, 11392) (11250 frames × 11392 pixels)  
 32 spatial × 356 spectral = 11392 pixels  
 Data: uint16 ADC counts, range 175-35961

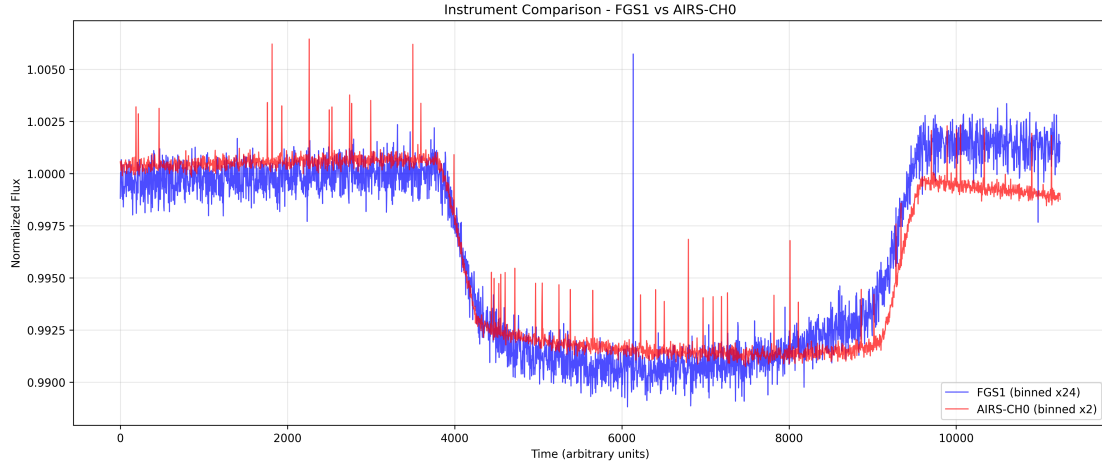


Figure 3.1.: unprocessed light curve binned for visualization

Additionally, each observed planet corresponds to an entry in the file `train_star_info.csv`, which describes the planet's properties:

- **planet\_id**: A unique identifier for the star-planet system
- **Rs**: Stellar radius in solar radii ( $R_{\odot}$ )
- **Ms**: Stellar mass in solar masses ( $M_{\odot}$ ), which together with **Rs** influences the scale of transit depth
- **Ts**: Stellar effective temperature in Kelvin, which affects limb-darkening coefficients
- **Mp**: Planetary mass in Jupiter masses ( $M_{\text{Jup}}$ )
- **e**: Orbital eccentricity (dimensionless)
- **P**: Orbital period in days
- **i**: Orbital inclination in degrees, which together with **e** and **P** influences transit shape and duration
- **sma**: Semi-major axis in stellar radii ( $R_s$ )



### 3.1.2. Targets

For each planet, the target output is the transit depth at 283 wavelengths (1 from FGS, 282 from AIRS). Transit depth is defined as  $\left(\frac{R_p}{R_s}\right)^2$  where  $R_p$  is the planetary radius including the atmosphere, and  $R_s$  is the stellar radius. This unitless ratio represents the fraction of stellar light blocked during transit. Atmospheric absorption causes wavelength-dependent variation in effective planetary radius - molecules such as  $H_2O$  and  $CO_2$  - absorb specific wavelengths, increasing the apparent  $R_p$  at those wavelengths and thus deepening the transit. The transmission spectrum (transit depth versus wavelength) encodes the atmospheric composition. The ground truth values for these transit are provided in the file `train.csv` representing the depth at each spectral bin.

```
train.csv:
- planet_id: Identifier
- wl_1 to wl_283: Transit depth at 283 wavelength bins
  Shape: (1100, 284)
```

## 3.2. Preprocessing

As is standard with data retrieved from sensors, each signal must be calibrated. The challenge issuers have provided an example calibration file<sup>2</sup> and it was decided to use these calibrations with minimal modifications. Early research showed that alternative approaches would demand significant domain knowledge beyond this project's scope. The signal calibration was done in seven steps described below. All illustrations were done using the planet 411903044.

1. **Analog-to-Digital Conversion (ADC)** - Photons collected by the detector generate photoelectrons that produce analog voltage signals proportional to the incident flux. The analog-to-digital converter quantizes these continuous voltage measurements into discrete digital counts to allow for computational analysis, while preserving the photometric information necessary to detect transit depth variations (see Figure 3.2). This conversion was executed using the file `adc_info.csv` which contains the gain and offset values for each sensor.
2. **Masking hot and dead pixels** - Photons can become oversaturated during the exposure, falsely reporting a strong signal even after transit. Alternatively, pixels can become dead and not record any photometry at any point during an observation. To avoid having these extreme values influence the transit depth analysis, these pixels were masked (see Figure 3.3). Contrary to the provided calibration function, the code uses `astropy.stats.sigma_clip` to replace these pixels with `nan` using the information from `dead.parquet` for each observation.

---

<sup>2</sup><https://www.kaggle.com/code/gordonyip/update-calibrating-and-binning-astronomical-data>

3. **Non-linearity correction** - In an ideal case, doubling the incoming photon flux for a pixel would double the ADC readout. In reality, pixel capacitors leak charge during integration, causing a response compression, or in simpler terms: bright signals read lower than they should. This distortion is deterministic and modelled via polynomial coefficients (see Figure 3.4). The provided calibration file `linear_corr.parquet` contains inverse polynomial coefficients that map the compressed ADC values back to true photon counts, restoring linearity before further analysis.
4. **Subtract dark current** - Even in complete darkness, detector pixels accumulate electrons from thermal energy in the semiconductor lattice. This dark current increases linearly with exposure time and varies spatially across the detector due to manufacturing imperfections and temperature gradients. The provided `dark.parquet` captures this pixel-specific baseline by imaging the detector under minimal exposure. Subtracting the dark current (scaled by integration time) isolates the photon signal from the thermal noise (see Figure 3.5).

$$corrected\_image = raw\_image - dark\_map \times \Delta t$$

5. **Correlated Double-Sampling (CDS)** - Each image is captured twice: once at exposure start and once at exposure end. The raw data alternates between these start and end frames. Applying CDS eliminates fixed-pattern noises and mitigates readout offsets by subtracting the initial state from the final state:

$$cds\_frame = frame\_end - frame\_start$$

The differential measurement cancels pixel-level biases present in both reads, such as reset noise, amplifier offsets, while preserving only the photon-induced charge accumulated during integration (see Figure 3.6). The alternating frame structure requires pairing: frames indexed with timestamps must be correctly matched before subtraction.

6. **Binning** - To reduce computational load and memory requirements, frames were temporarily binned by averaging consecutive observations. This strategy can also be used to align both sensors to the same temporal resolution, enabling synchronized analysis of their signals.
7. **Flat Fielding** - Not all pixels convert photons to electrons with equal efficiency. Quantum Efficiency (QE) varies across the detector due to manufacturing tolerances, dust, and optical path differences. A flat field is generated by uniformly illuminating the detector and recording the response map. Dividing science frames by this normalized flat field compensates for pixel-to-pixel sensitivity variations (see Figure 3.7):

$$corrected\_frame = \frac{cds\_frame}{flat\_field}$$

Without flat fielding, a spatially uniform astronomical source would appear mottled, introducing false structure into extracted spectra.

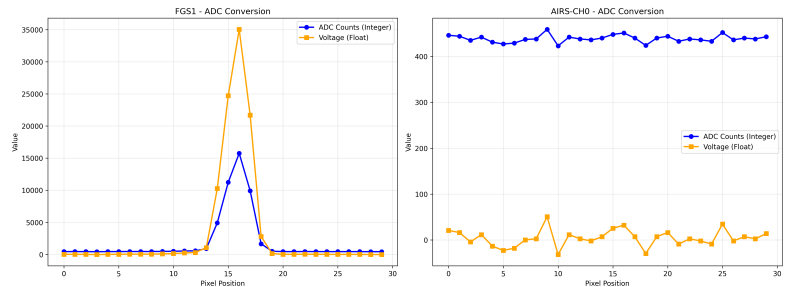


Figure 3.2.: ADC conversion for pixel 15 across entire observation.

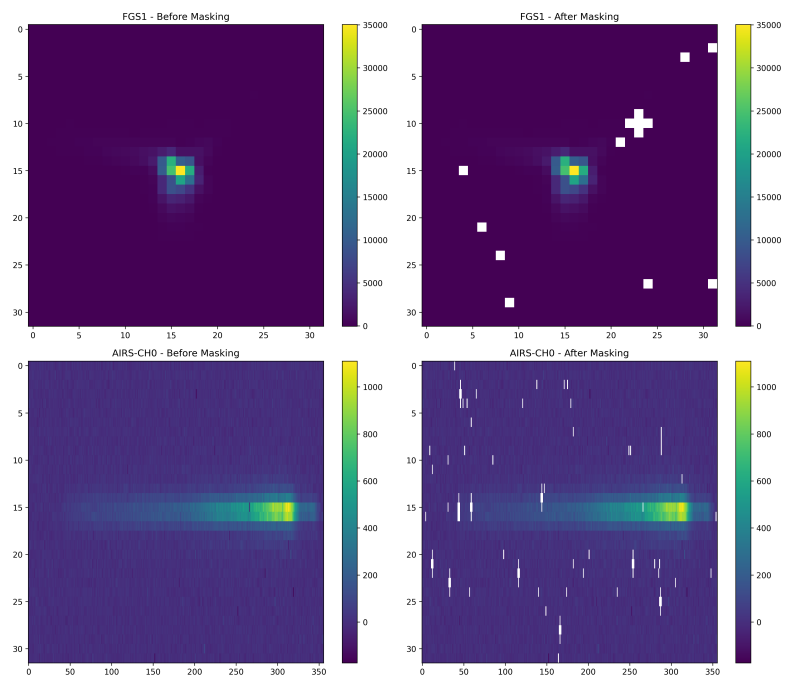


Figure 3.3.: Comparison of observation image without and with masked pixels.

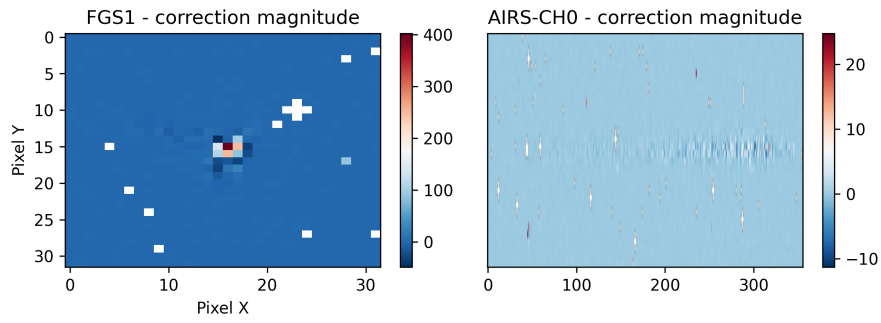


Figure 3.4.: Magnitude of correction across all pixels.

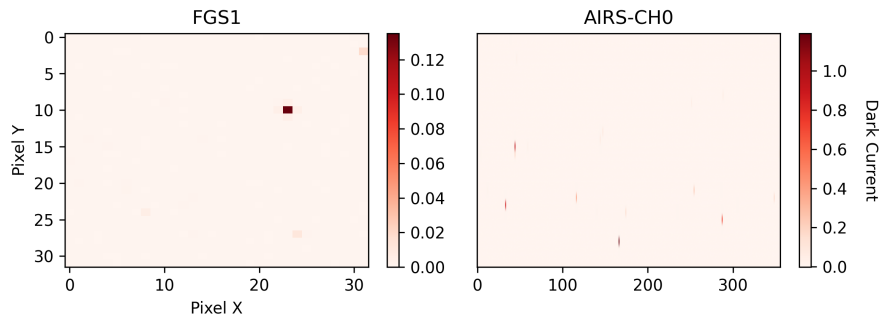


Figure 3.5.: Pixel map of dark current to be extracted.

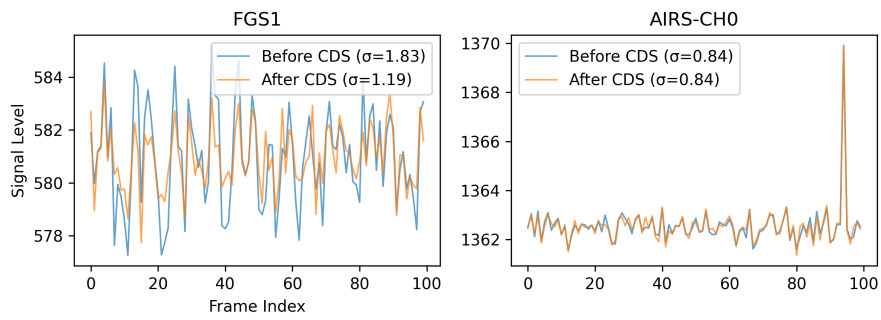


Figure 3.6.: Comparison with and without correlated double sampling.

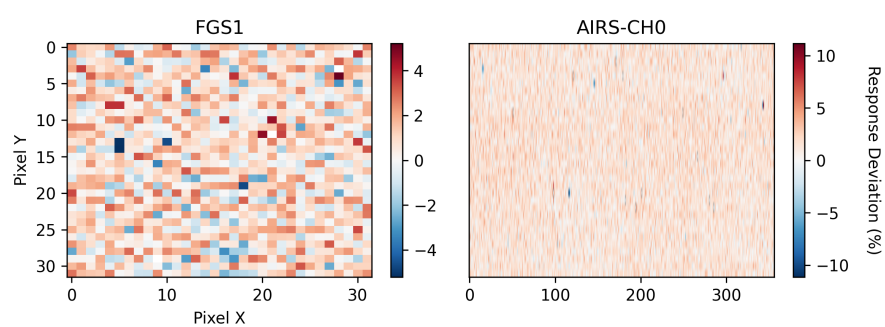


Figure 3.7.: Pixel deviation during uniform illumination.

## 4. Modeling

### 4.1. Models

#### 4.1.1. Preprocessing Pipeline

As mentioned in Chapter 3, the preprocessing pipeline provided by the challenge organizers was adopted with minimal modifications. The original implementation was systematically refactored to leverage `torch` tensor operations in place of `Numpy` arrays, yielding a computational speedup of approximately 40% across the entire calibration pipeline. This performance enhancement proved essential given the volume of data requiring processing during training and deployment on the Kaggle platform.

The binning strategy warranted careful consideration, as it represents a fundamental trade-off between temporal resolution and signal-to-noise ratio. While numerous approaches exist in the literature for optimizing bin sizes, it was ultimately decided to use a binning factor of 24 for FGS, and a factor of 2 for AIRS-CH0, resulting in 2812 temporal bins per sensor, which automatically aligned the sensor-observations as well.

A final preprocessing step involved the implementation of a clamping function to handle residual outliers that survived the `sigma_clip` applied during the hot pixel correction phase. Without this safeguard, extreme values which could have arisen from cosmic ray strikes or edge cases in calibration coefficients, might have dominated the dynamic range and obscured the subtle variation indicative of atmospheric transmission features. This preprocessing framework combines computational efficiency with robust albeit conservative outlier mitigation, ensuring that the downstream transit analysis operates on clean, well-conditioned photometric time series.

#### De-jitter algorithm

As described in the Problem statement (see section 1.1, being able to reliably identify and extract noise from the signal is crucial for effective signal processing. To enhance the provided preprocessing pipeline, we attempted to implement a de-jitter algorithm based on the paper of Bocchieri, A., Mugnai, L.V., Pascale, E. et al. (2025) presented in Chapter 2. The preprocessing from Section 4.1.1 was applied in the same sequence, with one modification to the binning procedure. A reduced binning factor of three was used to retain more signal information. Following the recommendations in the reference paper, flat fielding was applied, as preliminary tests showed it had no adverse effects. This modified algorithm was then applied to all FGS files in the dataset.

First, the frames with the highest SNR were identified, and their barycentre (or center of mass) were computed. The median of these positions was used as the reference position,

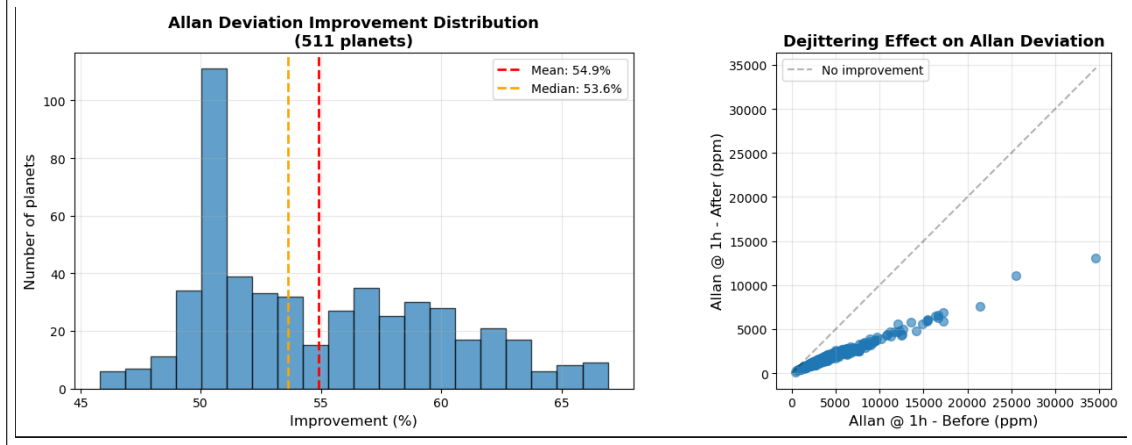


Figure 4.1.: Results of the de-jitter algorithm applied on the data set (FGS).

chosen for its robustness against outliers compared to the mean. Next, a vertical 1D cross-section through the spatial domain was analyzed to extract centroids, with the reference position serving as a first initial guess. The 1D IWCoG method was applied iteratively, either until convergence within a tolerance of  $1e-4$  or for a maximum of 128 iterations, whichever came first. This procedure returned the centroid position and width. Subsequently, higher-order moments were calculated with this centroid, returning its position, width, skewness and kurtosis. The entire process was then repeated horizontally. With the PSF characteristics determined, stellar parameters, including the star radius, semi-major axis, and eccentricity were extracted. The mid-transit time and transit depth were then estimated by locating the minimum flux point during the transit event.

To model the combined effect of jitter and planetary transit, we defined a composite model as the product of the jitter function and the transit model. In the fitting procedure, BATMAN computed the transit light curve for each set of trial parameters, while the jitter function was calculated from the extracted PSF moments. A non-linear least squares optimizer from the Scipy library iteratively refined all parameters until the combined model converged to the best fit with the observed flux. Once the optimal parameters were determined, the jitter component was reconstructed and removed from the observed flux by division, yielding the corrected light curve. When applied to the full dataset of 1 100 planets, the de-jitter algorithm successfully reduced jitter noise for 511 targets, achieving an average noise reduction of 54.9% in the FGS data (see Figure 4.1). For the remaining targets, noise levels remained moderate after correction, and crucially, no targets showed increased noise after processing. Despite these promising results, the algorithm was ultimately not incorporated into the final pipeline. The primary reason was inconsistent performance across the dataset — while effective for roughly half of the targets, the algorithm did not provide uniform improvement. Furthermore, as the project approached its deadline, insufficient time remained to conduct systematic comparative studies, such as training models both with and without de-jitter correction to quantify the impact on final detection performance.

## 4.2. Baseline model

Standard time-series approaches such as pure RNNs and LSTMs were deemed not suitable for this task: RNNs optimize for temporal dependencies - learning when events occur - but transit extraction requires understanding how the signal deviates from baseline across the entire observation window. Additionally, for an RNN-approach to work on a computational level, the transits would have to have been extracted with astrophysics libraries (e.g. BATMAN). This would have required accurate orbital parameters and would have risked truncating ingress/egress phases where atmospheric features manifest. Even more critically, isolating the transits from the rest of the signal removes context: the model must learn that the signal is noise-dominated even during transit to properly weight its predictions. Extracting the transits from their surrounding noise could therefore have resulted in the exact opposite. Instead, the problem was treated as a 1D computer vision problem. Convolutional filters perform edge detection across the full time series, identifying transit boundaries and depth variations without prior segmentation. This approach preserved global context while capturing local features at multiple timescales.

### 4.2.1. Architecture

The model comprises three parallel encoding heads that merge into a shared regression network: the FGS head (photometry), AIRS head(spectrometry), and the stellar parameter head.

The FGS head consisted of an input of a single channel photometric time series of shape: batch size, 1 and 135 000 divided by the binning factor. Its convolutional layers had 4 blocks (see 4.2) with decreasing kernel sizes (15-11-7-5) and stride 2 for downsampling. The larger initial layer allows to capture the full transit duration ( $\sim 3\,000$  frames) while the decreasing kernels detect finer structures such as limb darkening and stellar variability. Each block follows the convention sequence Conv1D, LayerNorm, ELU-activation. Dual pooling is applied, Adaptive Max Pooling to preserve the sharp absorption edges and Adaptive Average Pooling to capture the baseline transit depth. After passing through a fully connected layer, layer normalization and the ELU activation function, the output has 256 features.

The AIRS head's input is a parallel time series with 282 spectral channels which allows the network to learn correlated features (e.g. molecular absorption bands that span multiple channels). Its shape is: batch size, 282, 11 250 divided by the binning factor. 3 blocks of convolutional layers with decreasing kernel sizes (11 - 7 - 5) and downsampling with a size of 2 were implemented. The same steps are applied as in the first head: 1D convolution, layer normalization and ELU-activation. Dual pooling is applied as well and the head's output ends up with 512 features after passing through a fully connected layer, layer normalization and ELU. Table 4.1 summarizes the architecture of both sensors' heads.

Finally, the stellar parameters head provides the physical priors for interpreting the photometric/spectroscopic signals. Its input is of shape: batch size, 8, corresponding to all 8 stellar parameters. It passes through fully connected layers (8 - 64 - 128) with layer normalization and leaky ReLU (see Table 4.2).

A fusion network brings all three heads together. It normalizes across features within



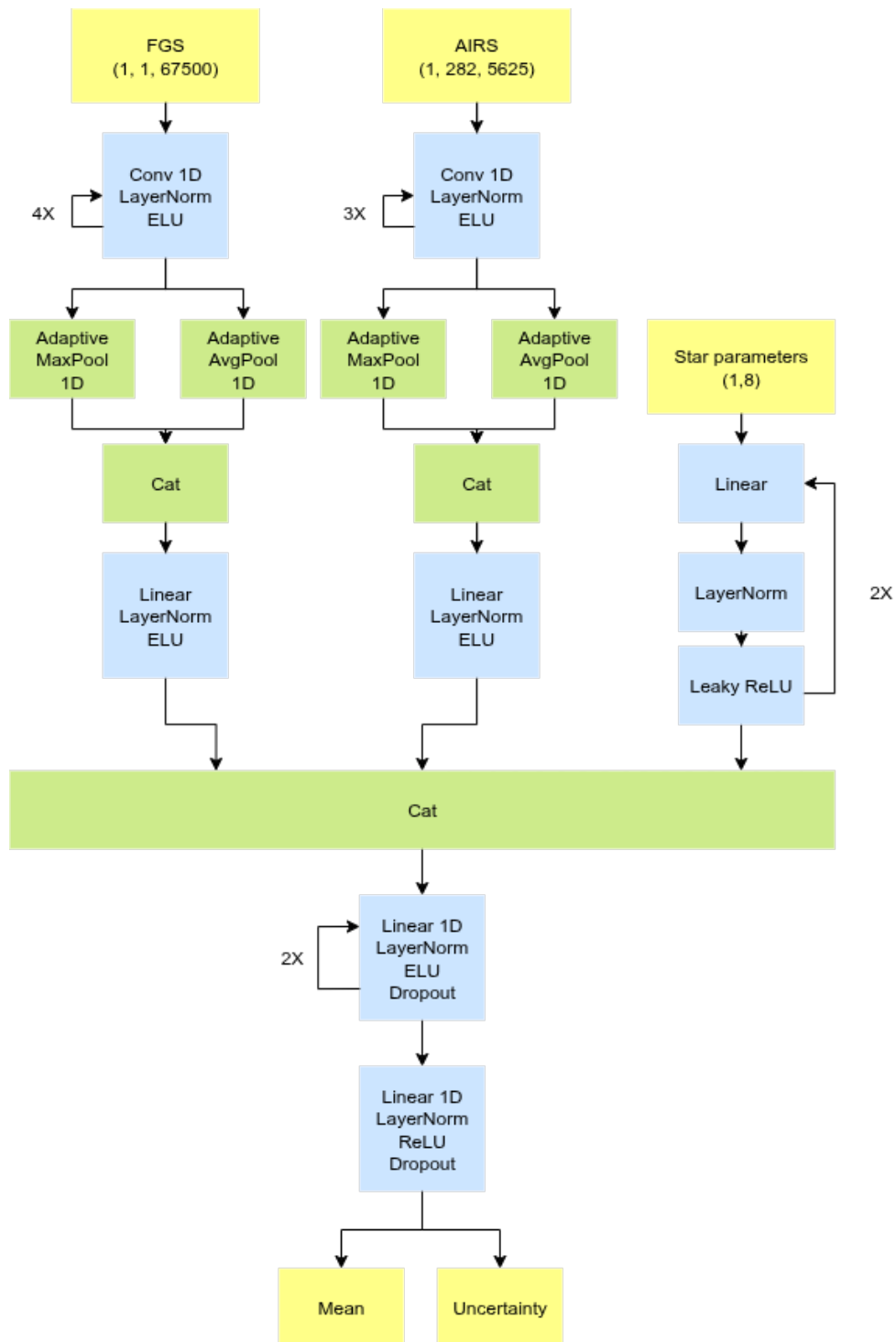


Figure 4.2.: Architecture of the baseline CNN model.

| Head                 | FGS                                            | AIRS                                           |
|----------------------|------------------------------------------------|------------------------------------------------|
| Input size           | batch,1,135 000/binning factor                 | batch,282,11 250/binning factor                |
| Convolutional layers | 4 blocks                                       | 3 blocks                                       |
| Channels             | 1 - 32 - 64 - 128 - 128                        | 282 - 256 - 256 - 256                          |
| Kernel size          | (15-11-7-5)                                    | (11 - 7 - 5)                                   |
| Stride               | 2                                              | 2                                              |
| Order                | Conv1D → LayerNorm → ELU                       | Conv1D → LayerNorm → ELU                       |
| Pooling              | Adaptive Max Pooling, Adaptive Average Pooling | Adaptive Max Pooling, Adaptive Average Pooling |
| Output features      | 256                                            | 512                                            |

Table 4.1.: Summary of the two first heads.

| Stellar parameters     | [Rs, Ms, Ts, Mp, e, P, sma, i] |
|------------------------|--------------------------------|
| Input                  | (batch, 8)                     |
| Fully connected layers | (8 - 64 - 128)                 |
| Order                  | FC → LayerNorm → LeakyReLU     |
| Output features        | 128                            |

Table 4.2.: Summary of the stellar parameters head.

each sample, which is critical with a high batch diversity where stellar types and orbital configurations are mixed. The ELU activation smooths gradients for negative values, improving convergence compared to ReLU for regression tasks. No dropout and no softmax/sigmoid are applied in the baseline model. The output is an unbounded regression with transit depths between 0.01 and 0.05. In total, we counted about 2.1 millions parameters.

#### 4.2.2. Improved CNN

To improve the baseline model, which performed abysmally with a Kaggle score of 0, several improvements were implemented. In a first step the Gaussian-Log-Likelihood function - used for the scoring on Kaggle (see Section 4.4.1) - was removed as loss function and only kept to contextualize the achieved results. The artificial clamping to values between 0 and 1 was removed for training and monitoring purposes. In a second step the K-fold cross-validation was implemented, allowing to use the fold disagreement to estimate the uncertainty (see Section 4.4.1).

#### 4.2.3. CNN - RNN

While RNN was not suitable for the full challenge, the data being sequential, it was considered to complement the CNN and for being compatible with the existing pipeline. Building on the baseline model, the data was first passed through three already implemented CNN layers. This served two purposes: learn some features and reduce dimensionality. Using

only RNN-layers would saturate the memory, and since RNNs pass through the entire sequence before returning an output, it could not be parallelized (Mohammadi et al., 2019). RNNs contain hidden layers, which receive the output of the previous layer and the input of the current timestep (Mohammadi et al., 2019). The sequence was originally an image, meaning looking into the future of the sequence did not leak information and allowing for a bidirectional model. This means the network maintained two hidden layers - one per direction - combined in the end into a single output (Mohammadi et al., 2019). The created model contained one RNN layer consisting of a Gated Recurrent Units (GRU). While RNNs should be able to *remember* long-term dependencies, older information tends to be lost. GRU use *gates* to decide what information should be remembered or forgotten. Its design should make it more robust to long-term dependencies. For similar performance, its computation should be more efficient than Long-Short-Term-Memories (LSTM). The built architecture was an adaptation of the baseline model. Both sensors were separately passed through three layers of 1D CNN (CNN, layer normalization, ELU activation function). The output was passed through the bidirectional GRU. The task to achieve was not a sequence-to-sequence nor a classification task. The mean and uncertainty had to be returned. Thus, the same steps as the baseline models were reused with the RNN output: adaptive max and average pooling, concatenation, linear fully connected layer, layer normalization and ELU activation function. While these steps were unusual for RNN, especially the pooling, they were the only ones found to deliver the desired outputs. After inputs from both sensors went through this pipeline, they were concatenated with the trained star parameters and trained again on a linear fully connected layer, with dropout, returning the mean and uncertainty. The detailed pipeline can be found on Figure 4.3.

#### 4.2.4. CNN - TCN

TCN were another variation of the baseline model, and considering sequentiality. A particularity of TCN was to have the same length for inputs and outputs. In addition, causal dilated convolutions are present, meaning an output at time  $t$  was only convolved with elements from time  $t$  and earlier in the previous layer. This prevented data leakage but was not critical for this use case. Dilated convolutions increase the receptive field and functions more or less like a context window, helping to control the memory size of the retained information. Theoretically, TCN should outperform LSTM and GRU (Bai et al., 2018). The built architecture is the same as the RNN one, except the TCN replaced the RNN layer (see Figure 4.4). It was implemented with the library PyTorch-TCN.

### 4.3. Model Training

#### 4.3.1. Training pipeline

Several preprocessed versions of the dataset were created early on to accelerate training process: one with steps 1 to 5 of the preprocessing pipeline (see Section 4.1.1), one with added binning, and one with added flat fielding, harnessing the entire preprocessing

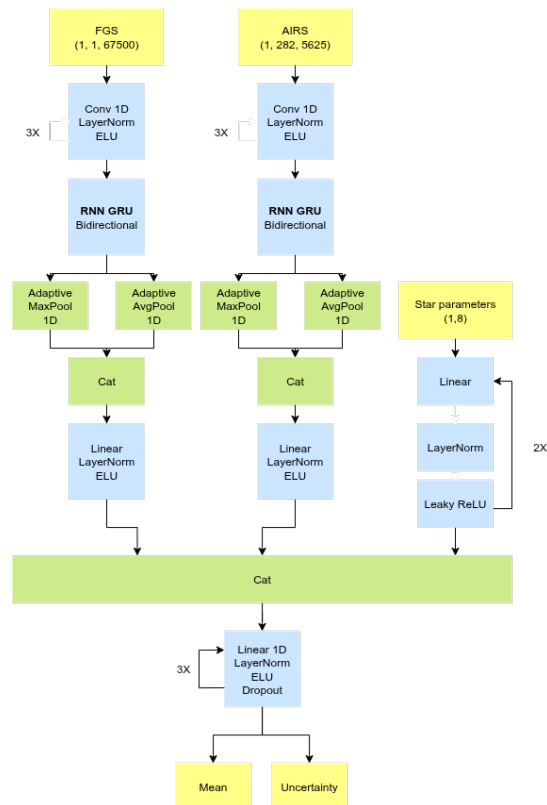


Figure 4.3.: Architecture of the CRNN model.

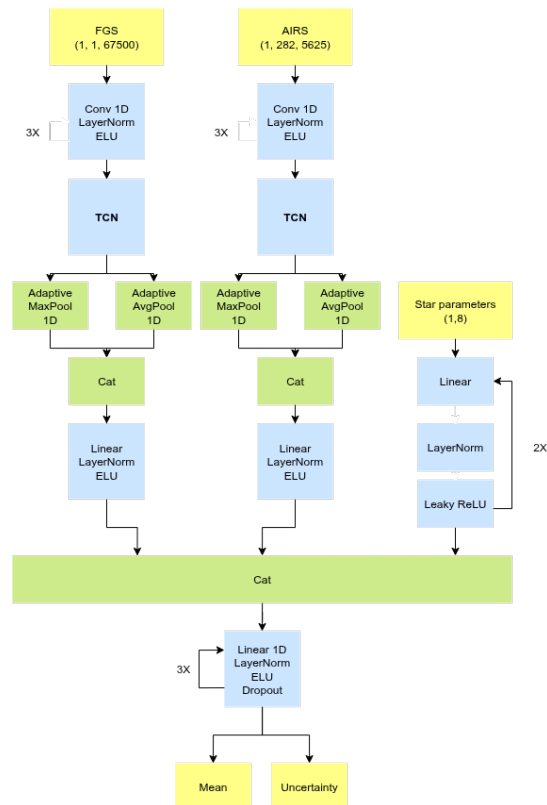


Figure 4.4.: Architecture of the TCN model.

pipeline.

The training pipeline integrated K-fold cross-validation with a 80/20 split directly, as uncertainty estimation was required by the challenge metric (see Section 4.4.1). Each fold was trained independently on its respective data split, producing  $k$  distinct model predictions per wavelength. The ensemble mean across folds provided the final prediction, while the standard deviation quantified the uncertainty estimate. An early stopping mechanism with a patience of 100 epochs was used to prevent overfitting, while still allowing for wild swings that unavoidably happened early in the training. Models trained for a maximum of 1 500 epochs typically converging between epochs 800 and 1 450 depending on the learning rate.

### 4.3.2. Experiments

The sheer volume of the training dataset (250GB) imposed severe computational constraints. As explained previously (see Section 4.3.1) the training necessitated several K-folds to predict uncertainty. The early intuition was to use a smaller subset of the data for hyperparameter experimentation. However, the results of these runs were not representative: under-performing combinations on the subset (Root Mean Square Error (RMSE):  $\sim 4000$  ppm) would produce some of the best results on the full dataset (RMSE:  $\sim 150$  ppm) or would show no improvement at all. This can most likely be explained with the assumed nature of the planetary transits. Initial data exploration showed a wide range within the wavelengths, suggesting that - as will happen during the real mission - the signal strength will vary greatly based on the type of host star in front of which the transit will happen and the distance between said host star and the probe.

Training on subsets therefore made the pipeline extremely vulnerable to excluding some of the more extreme or unique values, and to overfitting to a specific range. It was thus decided to use the full dataset for experimentation too, instead of extracting observations based on their signal strength. This slowed down progress during this phase significantly, but was ultimately deemed the more sustainable approach.

Using already preprocessed data, the resulting training run lasts about 1.5 hours per fold  $k$ . Unfortunately, experimentation was further hindered by the irregular time-out behavior of the HSLU-GPUhub, even during the allocated time slot for the module AICOMP. We experienced a lot of forced session-timeouts, which rendered the runs useless, and produced corrupted or empty Weights&Biases tracking files, leaving us with no information about the run's performance. This happened at such a frequent rate, that it could take up to two days for a successful run. For these aforementioned reasons, a systematic approach to hyperparameter tuning was all but impossible. Instead, the experimental efforts focused on identifying the optimal learning rate, the minimum number of folds required for stable uncertainty estimates, and normalization methods, which were launched and monitored manually. Thus, outlier folds handling, consistent decrease in validation loss, and more fine-tuning techniques were left out.

## 4.4. Model Evaluation

Model performance was assessed using metrics that reflect both the accuracy of transit depth predictions and the reliability of uncertainty estimates. Unlike traditional regression tasks where only point predictions matter, exoplanet atmospheric characterization requires probabilistic outputs, stating the model's confidence in its own predictions.

### 4.4.1. Loss function

The challenge requires predicting both mean transit depths and associated uncertainties for each wavelength. In signal processing, absolute values alone are insufficient; no matter how thorough the preprocessing, residual noise remains in the signal. The model must therefore report confidence in its predictions, distinguishing systematic model error from stochastic measurement noise. The model was trained using RMSE as the loss function to optimize mean predictions.

$$loss_{RMSE} = \sqrt{\sum \frac{(y - \mu)^2}{N}}$$

where  $y$  is the ground truth,  $\mu$  the prediction and  $N$  the number of samples.

Uncertainty estimation was handled through K-fold cross-validation:  $k$  independent models were trained on different data splits, producing  $k$  predictions per wavelength for each test planet.

The challenge evaluates both predicted spectra  $\mu_{user}$  and their uncertainties  $\sigma_{user}$  against the ground truth  $y$  across wavelengths using the Gaussian-Log-Likelihood<sup>1</sup>:

$$GLL = 0.5 \left[ \log(2\pi) + \log(\sigma_{user}^2) - \left( \frac{y - \mu_{user}}{\sigma_{user}^2} \right)^2 \right]$$

summed across all 283 wavelengths and all test planets. In general, the first part of the equation penalizes large variance, while the second estimates how close the predictions cluster around the ground truth. It penalizes both inaccurate predictions and miscalibrated uncertainties:

- underestimated  $\sigma$  : the second term explodes when prediction error exceeds reported confidence, heavily penalizing overconfident models.
- overestimated  $\sigma$ : the first term penalizes excessively broad error bars, discouraging models from hedging with uninformative predictions.

The ensemble mean from the K-fold cross-validation served as  $\mu_{user}$ , and the standard deviation across folds served as  $\sigma_{user}$ . This approach treated fold disagreement as a proxy for epistemic uncertainty. When folds converged to similar predictions, the model was confident, high disagreement indicated ambiguous cases or distribution shift.

The use of Gaussian Log-likelihood (GLL) reveals several expectations about the predictions. The metric assumed the correct errors ( $y - \mu_{user}$ ) follow a normal distribution.

---

<sup>1</sup><https://www.kaggle.com/code/metric/ariel-gaussian-log-likelihood>

This has implications for the heteroskedasticity within the errors: strong signals should have larger absolute errors than dim signals, following photon statistics where noise scaled with  $\sqrt{\text{signal}}$ . The Gaussian assumption also required stabilizing variance across folds through normalization and denormalization to prevent bright signals from overshadowing dim ones by accidentally capturing photon noise rather than model uncertainty.

### Scoring Metric

The official Kaggle scores on the leaderboard<sup>2</sup> used raw GLL values that were normalized to a [0,1] score via:

$$\text{score} = \frac{L - L_{ref}}{L_{ideal} - L_{ref}}$$

where:

- $L$  is the model's GLL on hidden test set.
- $L_{ref}$  is the GLL using training set mean/variance as constant predictions for all test planets (naive baseline).
- $L_{ideal}$  is the GLL assuming perfect predictions of the hidden test set ( $\mu = y$ ) with uncertainties matching ARIEL mission requirements (1 ppm for FGS, 10 ppm for AIRS).

All scores below zero were clipped to zero (Yip et al., 2024, p.6) This normalization anchored performance between a trivial baseline and the physical measurement limit.

### Wavelength Weighting

The GLL was computed as a weighted sum across wavelength channels to reflect their relative importance for atmospheric characterization. FGS provided broad-band photometric anchoring crucial for baseline correction, receiving a weight of 57.846 (approximately double the weight of all AIRS channels combined). Each of the AIRS channels received unit weight, as they collectively encoded the spectroscopic information necessary for molecular identification.

#### 4.4.2. Results

The variations of the baseline model using RNN and TCN showed lower results during training than the improved CNN. A version of the CRNN scored 0.111 on Kaggle but was quickly outperformed by another CNN version. For that reason, the exploration of the CRNN and TCN was reduced and the focus remained on further fine-tuning the CNN model. The table below shows the most promising CNN model versions which were deployed on Kaggle for official scoring.

##### Default settings:

<sup>2</sup><https://www.kaggle.com/competitions/ariel-data-challenge-2025/leaderboard>



- number of planets: 1 100
- preprocessing: ADC-conversion, masking hot/dead pixels, non-linearity correction, dark current subtraction, correlated double sampling
- patience: 100 epochs
- batch size: 128
- optimizer: AdamW
- weight decay: 1e-5

| Model          | Baseline1 | Baseline2 | No GLL   | Multiple obs. | No normalization | No normalization |
|----------------|-----------|-----------|----------|---------------|------------------|------------------|
| Folds          | 2         | 2         | 4        | 4             | 10               | 10               |
| Binning        | N/A       | N/A       | 1/12     | 2/24          | 2/24             | 2/24             |
| Flatfielding   | N/A       | N/A       | N/A      | yes           | yes              | yes              |
| Learning rate  | 1e-3      | 1e-4      | 1e-3     | 1e-4          | 1e-4             | 1e-05            |
| Warmup         | N/A       | N/A       | N/A      | 10            | 10               | 10               |
| Epochs         | 1 000     | 1 000     | 1 000    | 1 500         | 1 500            | 1 500            |
| Early stop     | 999,913   | 861,896   | None     | 1 239-1 498   | 975-1 488        | 1 461-1 500      |
| Train loss     | -4.8896   | -4.3199   | -1.7196  | 2.66e-5       | 6.7e-4           | 6.2e-4           |
| Value loss     | -4.8657   | -4.2994   | -1.5695  | 3.31e-5       | 9.21e-4          | 11.77e-4         |
| RMSE-AIRS      | 16.4039   | 49.3441   | 115.5391 | 41.03         | 10.4078          | 32.9552          |
| Sigma-AIRS     | N/A       | N/A       | 159.356  | 101.24        | 9.8088           | 10.5687          |
| RMSE-FGS       | 27.7803   | 25.350    | 143.8669 | 4.468         | 10.1293          | 43.0876          |
| Sigma-FGS      | N/A       | N/A       | 118.9021 | 17.213        | 4.4223           | 5.1977           |
| Local score    | 0.6857    | 0.6183    | 0.712    | 0.9212        | 0.828            | 0.9379           |
| Official score | 0.000     | 0.000     | 0.111    | 0.115         | 0.283            | 0.278            |

It has to be noted that there are fundamental differences between the CNN models in Table 4.4.2; they are nonetheless all based on the initial baseline model. The Baseline models fed the RMSE directly into the GLL function which was used as the loss function. As FGS is weighted with a factor of 57, this did not encourage the model to correctly predict AIRS-values. For the 3<sup>rd</sup> model, the GLL was removed from model evaluation and only kept for context and monitoring, this was the first version to score with 0.111. The 4<sup>th</sup> model made use of the multiple observations that 10% of the planets contained, averaging their values in hope for more robust predictions, increasing the score by 0.004. The 5<sup>th</sup> model had all of its normalization logic removed. Previously, both the signals and the targets had a z-score normalization applied, as is standard. However, as we used the resulting mean and variance as constants to predict the values of the unknown, hidden test set, which contained unseen ranges, there was a chance of the normalization interfering with exactly that. The more than doubling of the score to 0.283 confirmed this strange hunch. No further solution was found from that point. Many attempts to tune hyperparameters or discard outlier folds were done, with no score improvement, as shown exemplary with the 6<sup>th</sup> version.

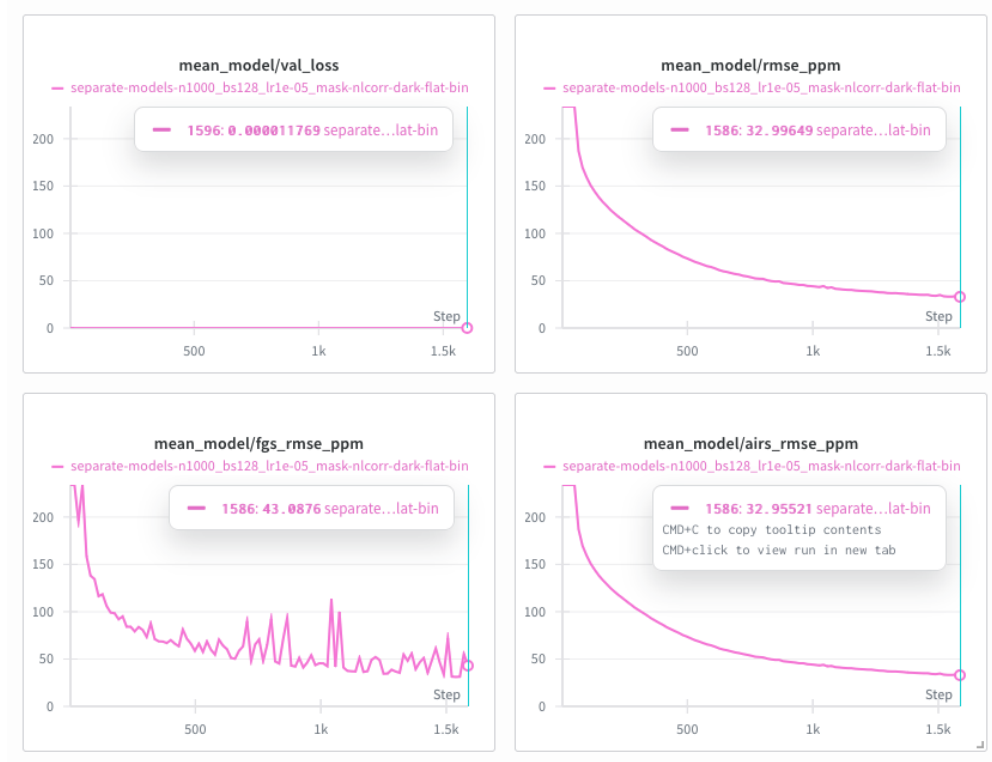


Figure 4.5.: Model v6 in Weights&Biases

## 4.5. Error Analysis and Improvements

The error analysis proved - as everything else in this project - to be very challenging. Despite having reverse engineered the scoring system (see Section 4.4.1), there was a clear disconnect between the the local Kaggle score, and the official scoring. While so far there has been no confirmation from the challenge organizers as to how much of the hidden test set was manipulated and in what sense (new host star type, stellar parameters, etc.) it can be safely assumed that between 10 and 20% of the hidden test set must have been affected (as was in previous years). This would imply, that the official score should experience a degradation of a similar magnitude compared to the reverse engineered local score, assuming some trouble with generalization which should be expected with such a project. However, the discrepancy was a lot bigger (50 to 60%). There are several potential explanations for this.

First, since the end of the competition, there have been several public discussions on Kaggle between high placing teams, about the hidden test set. Several of these teams appear to be from the domain of astrophysics and were wondering if the manipulation of the hidden test set had been too severe, with too less information given, for a model to

generalize<sup>34</sup>. Most notably, the winner of the competition himself admitted that his approach would have only resulted in a score of  $\sim 0.5$  without the application of what he called "severe fudging" which brought the score up to 0.648.

Second, the error analysis revealed a disconnect between the theoretical expectation and the empirical results observed using the GLL function. According to its formula, predictions with well-calibrated uncertainties that follow normal distribution, should achieve a superior score. In practice, the exact opposite was observed. In Table 4.3, Fold 1 performed closest to a Gaussian Distribution, but resulted in a negative score, compared to Fold 2 with the worst distribution but the highest score. This counterintuitive result stems from the asymmetric structure of the GLL: while the  $\log(\sigma^2)$  term penalizes large uncertainties, this penalty scales logarithmically. The  $\frac{(y-\mu)^2}{\sigma^2}$  term, which rewards prediction accuracy scales quadratically. Consequently, raw prediction accuracy dominates the score. This suggests that models trained on RMSE excel at minimizing prediction error on the training distribution, but the fold disagreement - which should penalize such overfitting through increased epistemic uncertainty - fails to influence the GLL evaluation meaningfully. The logarithmic penalty is insufficient to counteract the quadratic accuracy reward, rendering the uncertainty term effectively irrelevant and allowing overfitted models to achieve high scores locally. And of course then very low scores when confronted with new data on Kaggle.

| Metric                    | Fold 1  | Fold 2 | Fold 3 |
|---------------------------|---------|--------|--------|
| Kaggle Score              | -0.066  | 0.418  | 0.356  |
| FGS RMSE (ppm)            | 1063.34 | 610.31 | 579.65 |
| FGS MAE (ppm)             | 411.71  | 279.40 | 303.97 |
| FGS Mean $\sigma$ (ppm)   | 600.04  | 476.53 | 438.88 |
| FGS Within $1\sigma$ (%)  | 73.2    | 87.6   | 78.2   |
| FGS Within $2\sigma$ (%)  | 86.8    | 98.8   | 92.4   |
| FGS Within $3\sigma$ (%)  | 94.7    | 99.4   | 97.6   |
| AIRS RMSE (ppm)           | 1115.81 | 464.45 | 554.49 |
| AIRS MAE (ppm)            | 472.33  | 233.40 | 285.00 |
| AIRS Mean $\sigma$ (ppm)  | 591.91  | 438.23 | 447.30 |
| AIRS Within $1\sigma$ (%) | 70.5    | 89.2   | 81.0   |
| AIRS Within $2\sigma$ (%) | 88.8    | 98.5   | 95.0   |
| AIRS Within $3\sigma$ (%) | 93.9    | 99.5   | 98.2   |

Table 4.3.: Comparison of fold performance metrics with 110 planets.

Third, it is possible that keeping signals intact was not the best approach. As the competition had finished before this project started, during the course of this semester more and more participants revealed information about their approach. The consensus among the higher placing contestants appeared to be, that without carefully extracting the transits, no meaningful results can be generated. As already discussed in Chapter 3.2 this was a conscious decision taken early on from our side, to avoid spending even more time gather-

<sup>3</sup><https://www.kaggle.com/competitions/ariel-data-challenge-2025/discussion/609706>

<sup>4</sup><https://www.kaggle.com/competitions/ariel-data-challenge-2025/discussion/602425>

ing domain knowledge. Transit curve extraction is a delicate process, as some of the most meaningful information is hidden in the beginning and end of a transit. Without additional knowledge and experience it would have been all too easy to accidentally cut off this part, or keeping too much noise in, distorting the results beyond what we would have been able to analyze and mitigate.

Fourth and lastly, during the later period of the training there were hints that the implemented cross-validation, while necessary for the uncertainty, was too vulnerable to the split logic, especially the resulting distributions within the train and validation sets. It would have been counterproductive to reproduce a perfect imitation of the test set distribution for each split, but the naturally occurring variances in the splits might have overwhelmed the model. It was observed that several times, when a split was performing badly during validation, the training RMSE was actually within a good range. This is a clear indicator of a model learning the training distribution and then failing to generalize on validation. Which happened most likely because the signals in the test set were too different from the validation set.

## 5. Conclusion

This project was part of the Kaggle Competition NeurIPS Ariel Data Challenge. The aim was to develop a model that can predict the mean and uncertainty of 283 wavelengths during a planetary transit from 1 100 unseen, long, and noisy signals.

A multi-headed convolutional architecture was developed that processed photometric data from FGS, spectroscopic data from AIRS, and stellar parameters through parallel encoding streams before fusion. Several architectural variants were explored, such as RNN and TCN extensions to the baseline CNN. These did not lead to improved results, confirming the CNN was most suited for this challenge and sequentiality could be disregarded. A de-jitter algorithm based on PSF moment extraction was implemented and tested, but ultimately not integrated into the model, due to inconsistent performance across the dataset. The best performing model - an improvement to the baseline model - achieved a local score of 0.94 using the reverse-engineered GLL metric and an official Kaggle score of 0.28.

## 6. Reflection

This project presented a very big challenge for us. Based on our limited experience with engineering problems such as signal processing and calibrations, we decided early on to research the domain as much as possible, as we suspected without domain knowledge we would not get anywhere. Learning about space, machine learning methods, computer vision, time series and more, was very rewarding. However, despite all our research and preparedness, it was also a very frustrating experience. Astrophysics is not a domain one can simply google or query an LLM with. We did rely heavily on it during the initial research, but more often than not the answers were contradictory or downright nonsensical. It is a very niche and complex domain, and machine learning for astrophysics even more so. More often than not we found ourselves with results we couldn't interpret, found online discussions we could barely follow, and had a hard time finding support. Working in a domain where there are no best practices or established approaches yet, least a pretrained model, made it especially hard to orient ourselves and have confidence in the decisions taken during this semester. The additional computational limitation caused by the GPUHub only contributed to this stressful experience. The Hub was not suited for reliable training on such big data set.

Nonetheless, despite all the challenges, we managed to create our own architectures and implemented an algorithm we had to reverse-engineer based on information in a research paper. We learned a lot during this project, most importantly how to handle a problem, to which there is no solution yet. And in the end, after all the time spent we were rewarded with a score of 0.283 on Kaggle, which was very gratifying.

# Glossary

**ADC** Analog-to-Digital Conversion. 12

**AIRS** Ariel Infra-Red Spectrometer. 7, 8, 10, 12, 19, 21, 27, 32

**ARIEL** Atmospheric Remote-sensing Infrared Exoplanet Large-survey. 1, 2, 4, 6, 10, III, VIII, IX

**CDS** Correlated Double-Sampling. 13

**CNN** Convolutional Neural Network. 3, 4, 8, 21, 22, 27, 32, II

**dark current** The dark current is the current flowing through the detector in the absence of any incident photon flux. 13

**DNN** Deep Neural Network. 5

**eccentricity inclination** Angle between the orbit plane and the ecliptic plane. 7

**ESA** European Space Agency. 10

**exoplanet** Planet outside the Solar System. 1

**exosim** Exoplanet Observation Simulator. 6

**extrasolar planet** Exoplanet. 1, 2

**FGS** Fine Guidance System. 7, 8, 10, 12, 17–19, 21, 27, 32, III

**flat field** Response of the telescope-camera system to a source of uniform illumination; image of a uniformly illuminated field. 13

**GLL** Gaussian Log-likelihood. 26, 27, 30, 32

**GRU** Gated Recurrent Units. 22

**IWCoG** Iterative Weighted Center of Gravity. 7, 18

**jitter noise** Noise coming from the vibrations and drifts of the telescope. 2

**Kaggle** Kaggle is a competition platform with communities, tools and resources for data science projects. 3, 4, 21, 27, 29, 32, II

**limb darkening** Phenomenon of intensity decrease observed toward the limbs of the stars. 10

**LLM** Large Language Model. IV, XII

**LSTM** Long-Short-Term-Memories. 22

**MSE** Mean Squared Error. 5

**NDR** Non-Destructive Readout. 10

**orbital inclination** . 7

**PCA** Principal Component Analysis. 8

**period** The time it takes an orbiting body to make one complete revolution around the Sun. 7

**PSF** Point Spread Function. 6, 18, 32

**QE** Quantum Efficiency. 13

**RMSE** Root Mean Square Error. 25, 26

**RNN** Recurrent Neural Network. 3, 21, 22, 27, 32

**semi-major axis** One half of the major axis of the elliptical orbit; also the mean distance from the Sun. 7

**SNR** Signal-to-Noise Ratio. 4, 17

**stellar spot** A stellar spot, or sun spots are regions that appear darker because they are cooler by 1 000 or 2 000 degrees Celsius. The phenomenon is temporary. 4

**TCN** Temporal Convolutional Network. 3, 22, 27, 32

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## **A. Additional illustrations**

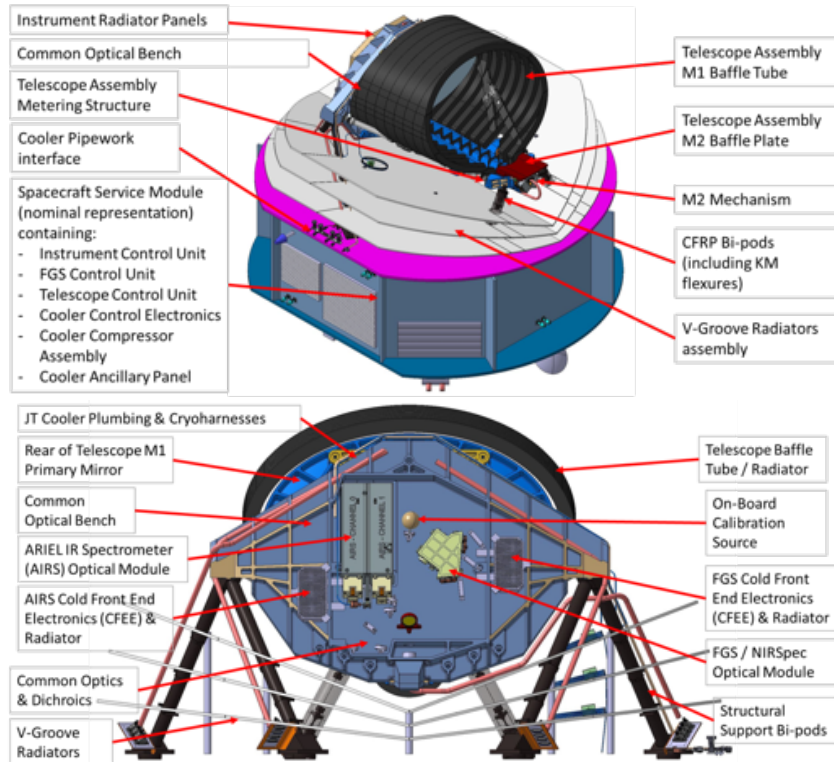


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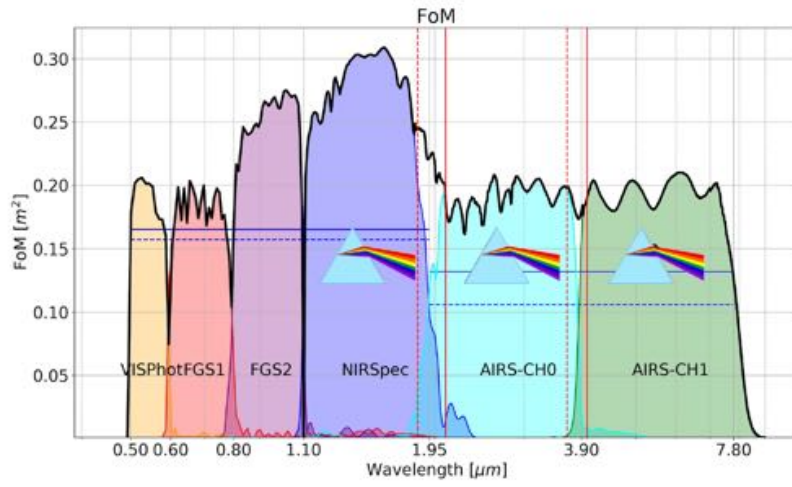


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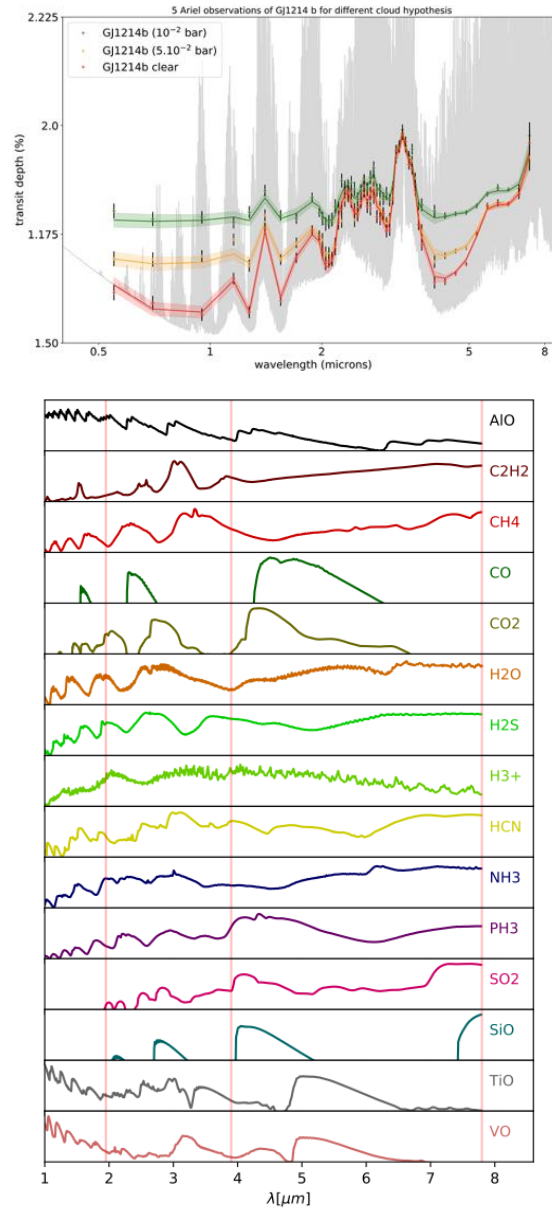


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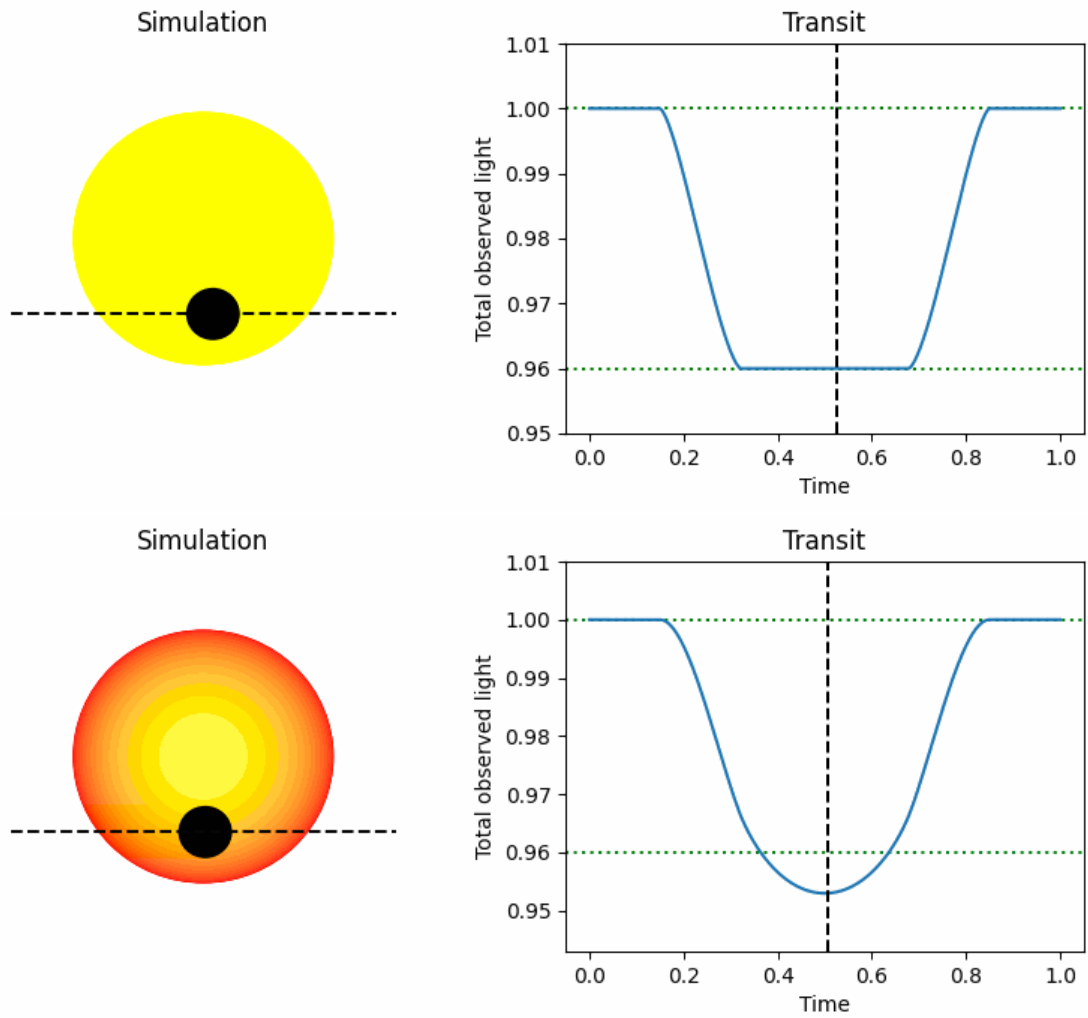


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## **B. Usage of LLM**

| Intended use       | Description of the specific use                                                        | Area                                 | AI tool(s) used                                                                                                                                                                                           |
|--------------------|----------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation     | Implement de-jitter algorithm, architecture models, and training/validation pipelines. | Scripts on Gitlab                    | ChatGPT <a href="https://chatgpt.com">https://chatgpt.com</a> ; Claude <a href="https://claude.ai">https://claude.ai</a> ; Perplexity <a href="https://www.perplexity.ai/">https://www.perplexity.ai/</a> |
| Translation        | Translation of words or sentences between French, German and English.                  | Literature search and report writing | DeepL <a href="https://www.deepl.com/en/translator">https://www.deepl.com/en/translator</a>                                                                                                               |
| Grammar            | Verify the grammar of sentences.                                                       | Report writing                       | Perplexity <a href="https://www.perplexity.ai/">https://www.perplexity.ai/</a>                                                                                                                            |
| LaTeX debugging    | Solve errors and warnings, create or modify tables, fix bugs of the template.          | LaTeX code of the report             | ChatGPT <a href="https://chatgpt.com">https://chatgpt.com</a> ; Claude <a href="https://claude.ai">https://claude.ai</a> ; Perplexity <a href="https://www.perplexity.ai/">https://www.perplexity.ai/</a> |
| GitHub fix         | Solve issues with GitHub actions such as merge, push.                                  | Code management                      | ChatGPT <a href="https://chatgpt.com">https://chatgpt.com</a>                                                                                                                                             |
| Data Visualization | Debug and adapt input plots                                                            | Visualization                        | Claude <a href="https://claude.ai">https://claude.ai</a>                                                                                                                                                  |
| Code Refactoring   | translate changes between local .py file and kaggle .ipynb file                        | Code management                      | Claude CoPilot                                                                                                                                                                                            |
| Code Fixes         | Debugging and logic flow corrections                                                   | Code management                      | Claude <a href="https://claude.ai">https://claude.ai</a>                                                                                                                                                  |

Table B.1.: Usage of AI tools